

# Session 7: CPSA formalism, and Beyond CPSA

Slides prepared by UMBC Protocol Analysis Lab

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- ▶ Proofs *about* theorem provers are extremely complicated
  - ▶ Important if you are interested in proving things *about* CPSA, rather than *with* CPSA.



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- ▶ Scenario + Assumptions + Set of actions possible = analysis
- ▶ “Enrichment” refers to finding plausible actions to explain the scenario
- ▶ Once the scenario is explained, then a skeleton is called *realized*

# Post Quantum Cryptography

What the future of crypto looks like, but when?

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Post-quantum attacks are not abstracted from the primitive, so CPSA cannot consider them.



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Grover's algorithm has, so far, not been shown to be feasible.



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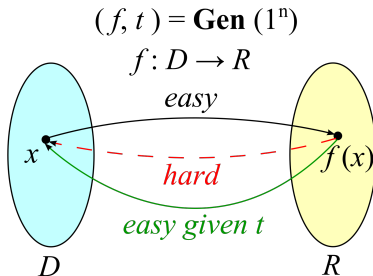
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- ▶ Lattice problem as trapdoor, average case NP-hard



**Figure:** By IkamuseFan - Own work, CC BY-SA 4.0,  
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- ▶ Symmetric key schemes require both parties to have the same key (hard)
- ▶ Symmetric key schemes can be vulnerable to other attacks

# Merkle signature scheme

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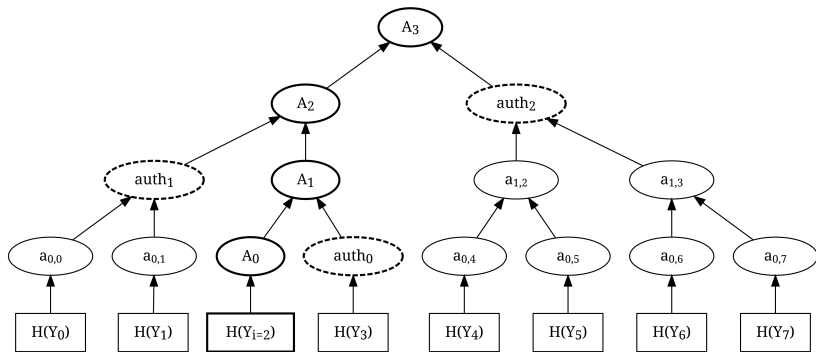
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**Figure:** By MaxLevy137 - Own work, CC BY-SA 4.0,  
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# Survey

Fill in details about the survey here!